

# Carrier 38MURA Parts List (2019)

Model 38MURA | 2019 | Parts

PL - 38MURA - 2019

Prototype sample document for the clickable demo. Not an official Carrier publication.

| Rev | Date    | Description                                   |
|-----|---------|-----------------------------------------------|
| 1.0 | 2019-05 | Initial release for the 2019 model year.      |
| 1.1 | 2019-09 | Updated CN2 pinout and fault-code table.      |
| 1.2 | 2020-02 | Added inverter diagnostics and torque values. |

For professional use only. Service work must be performed by qualified personnel.

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# Control Box Assembly

## Section 2: Control Box

The control box assembly contains the control board, low-voltage transformer, run capacitor, and motor harness. Order the control board by its full part number to match the 2019 revision.

**Control Box Assembly**

1 Control board

2 Transformer

3 Capacitor

4 Harness, CN2

**Callouts**

1. Control board

2. Transformer

3. Capacitor

4. Harness, CN2

Control box assembly callouts

| Item | Part Number    | Description                    |
|------|----------------|--------------------------------|
| 1    | CB-38MURA-2019 | Control board                  |
| 2    | XF-24VA-01     | Low-voltage transformer, 24VAC |
| 3    | CAP-35-370     | Run capacitor, 35/370          |
| 4    | HRN-CN2-01     | Blower motor harness, CN2      |

The control board is CB-38MURA-2019. Confirm the revision suffix before ordering.

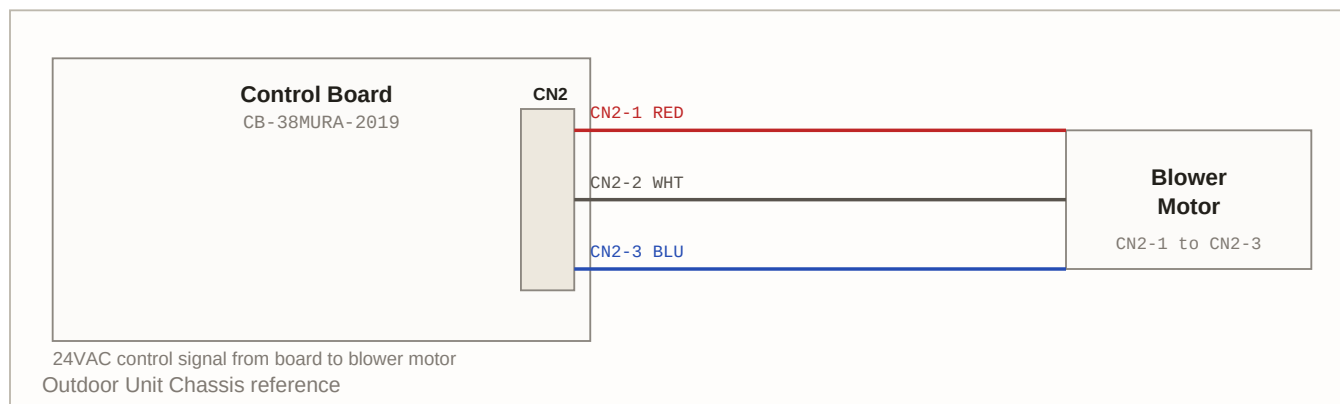
## 4. Outdoor Unit Chassis

Connector CN2 carries the low-voltage control signal between the board and the blower motor. A loose or corroded pin here is one of the most common causes of intermittent operation.

### Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

| Speed Tap | CFM  | Static     |
|-----------|------|------------|
| Low       | 600  | 0.20 in wc |
| Med-Low   | 800  | 0.30 in wc |
| Med-High  | 1000 | 0.40 in wc |
| High      | 1200 | 0.50 in wc |



Always confirm 0VAC and 0VDC at the board before making resistance measurements.

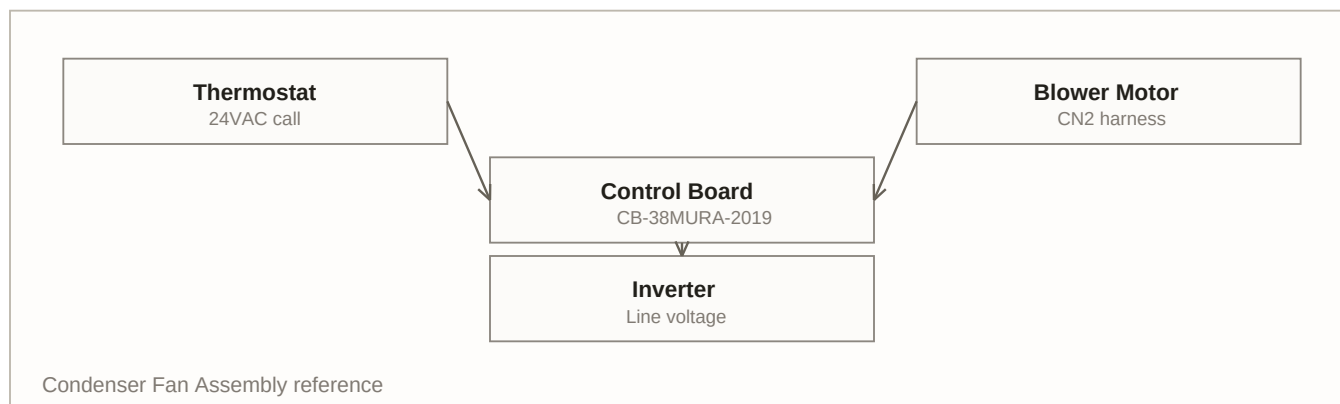
## 5. Condenser Fan Assembly

The board stores the last five fault codes in non-volatile memory. Review the stored codes before clearing them, because the history often points to the true root cause.

### Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

| Item         | Expected    | Notes                     |
|--------------|-------------|---------------------------|
| Line voltage | 208-230 VAC | +/-10%                    |
| Low voltage  | 24 VAC      | +/-10%                    |
| DC bus       | 310 VDC     | Discharged before service |
| CN2 signal   | 24 VAC      | Call for fan              |
| Ambient      | see chart   | Record at test            |



Record the outdoor ambient and the suction line temperature together for an accurate comparison.

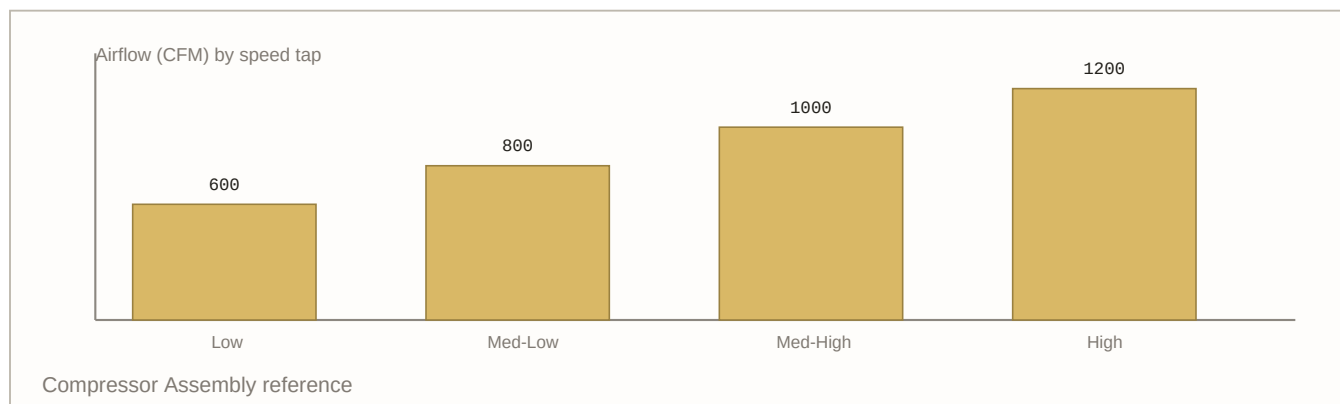
## 6. Compressor Assembly

Record ambient and supply-air temperatures at the same time so the temperature split can be compared against the charging chart for the current mode.

### Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

| Terminal | Wire | Function      |
|----------|------|---------------|
| CN2-1    | RED  | 24VAC supply  |
| CN2-2    | WHT  | Common return |
| CN2-3    | BLU  | Tach feedback |
| CN1-1    | BLK  | Earth ground  |
| CN1-2    | YEL  | Thermostat Y  |



If a code returns after clearing, inspect the harness before replacing the board.

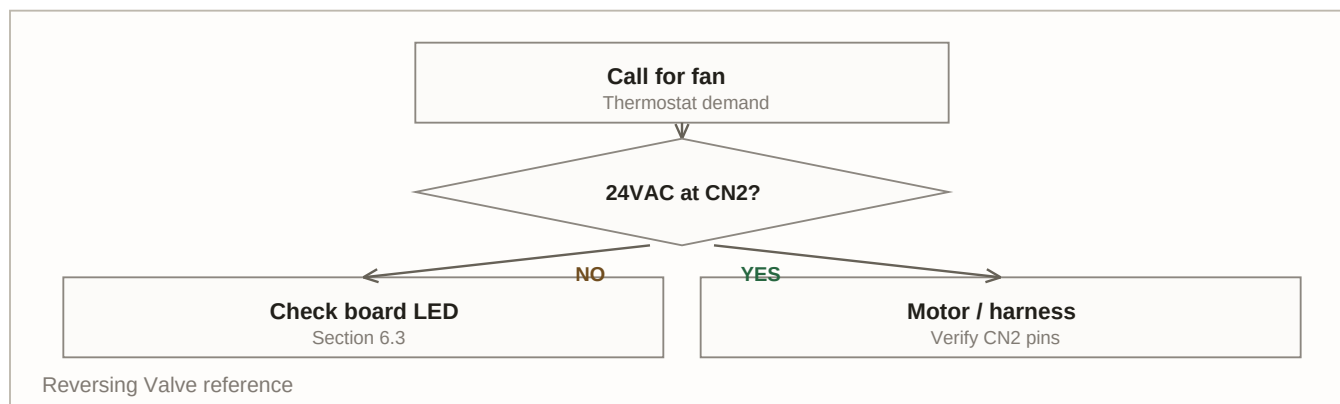
## 7. Reversing Valve

When a reading is out of range, do not replace the component until the associated wiring and connectors have been verified good. Most returned parts test good on the bench.

### Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

| Code | Meaning        | Action                   |
|------|----------------|--------------------------|
| E1   | High pressure  | Check charge and airflow |
| E2   | Low pressure   | Check charge and leak    |
| E4   | Locked rotor   | Check compressor         |
| E5   | AC overcurrent | See Section 5.2          |
| E6   | Communication  | Check harness            |



Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

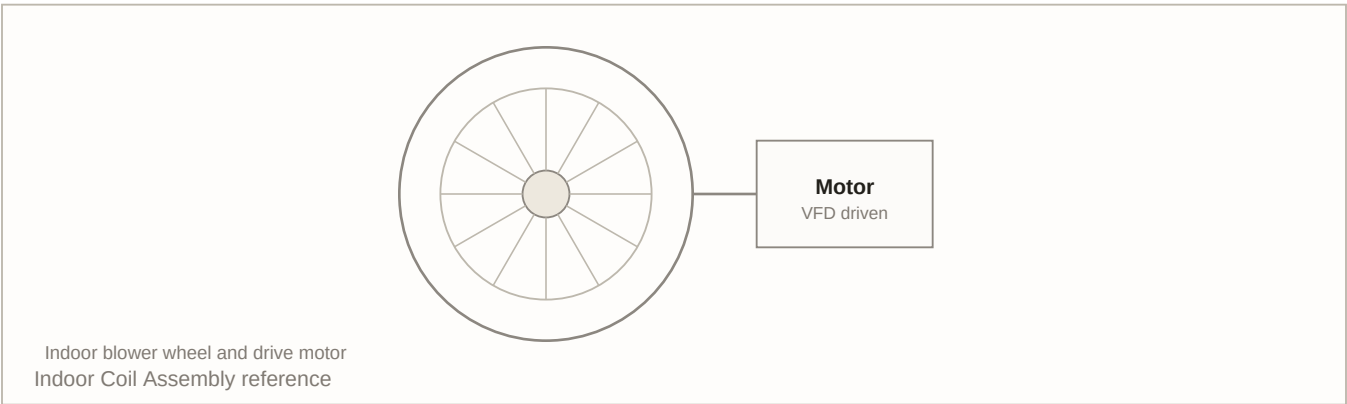
## 8. Indoor Coil Assembly

Torque all electrical terminations to the value shown in the specifications table. Loose terminations overheat and can damage the board terminals over time.

### Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

| Step            | Tool          | Spec      |
|-----------------|---------------|-----------|
| Mounting screws | 1/4 in driver | 2.5 N-m   |
| Terminals       | Torque driver | 0.8 N-m   |
| Line set        | Flare wrench  | per chart |
| Filter          | Hand          | snug      |
| Panel           | 1/4 in driver | 1.5 N-m   |



Always confirm 0VAC and 0VDC at the board before making resistance measurements.

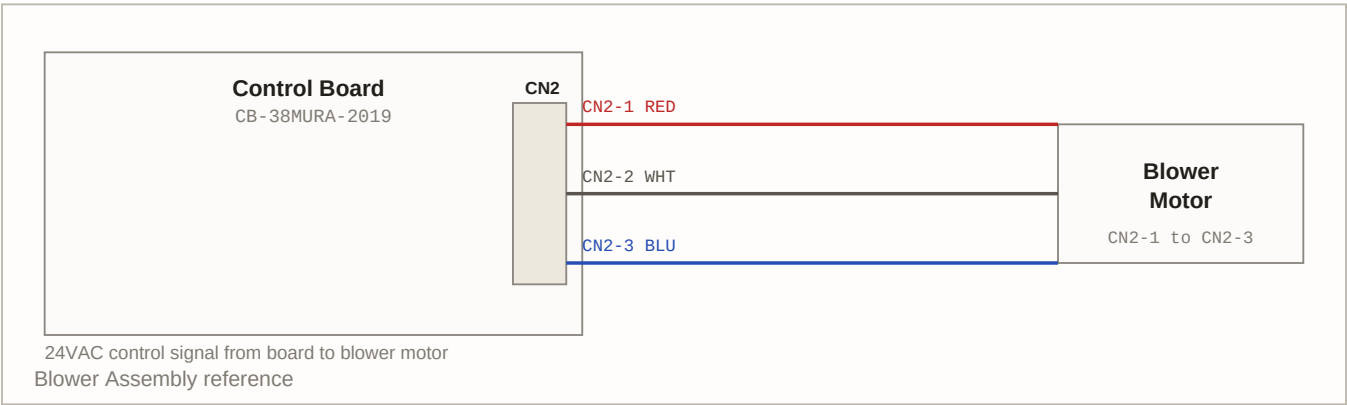
## 9. Blower Assembly

Document the final readings and the parts used on the work order. Accurate records shorten the next visit and support the warranty claim.

### Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

| Speed Tap | CFM  | Static     |
|-----------|------|------------|
| Low       | 600  | 0.20 in wc |
| Med-Low   | 800  | 0.30 in wc |
| Med-High  | 1000 | 0.40 in wc |
| High      | 1200 | 0.50 in wc |



Record the outdoor ambient and the suction line temperature together for an accurate comparison.

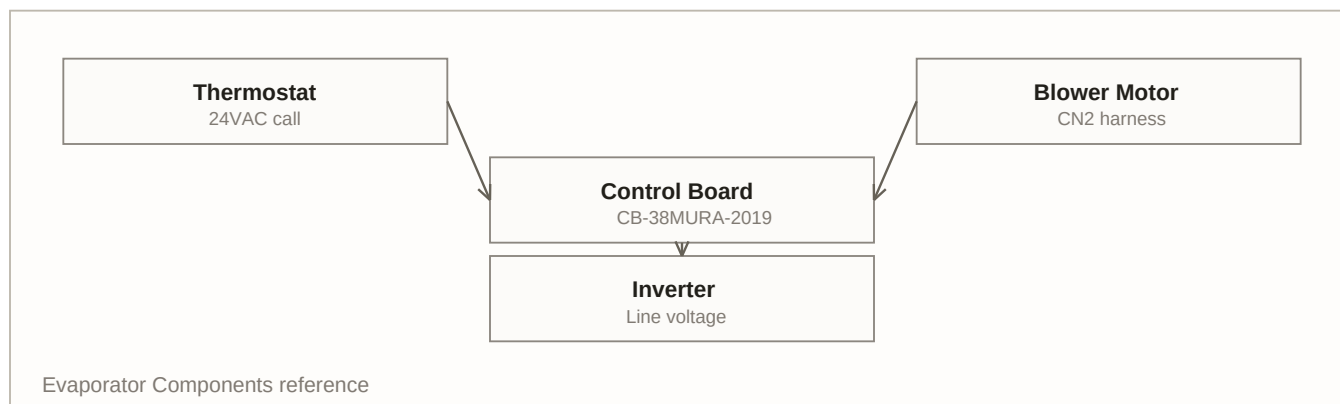
## 10. Evaporator Components

This section covers the recommended field procedure for the 2019 Carrier 38MURA. Follow the steps in order and verify each measurement before moving to the next step. Record every reading on the service worksheet.

### Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

| Item         | Expected    | Notes                     |
|--------------|-------------|---------------------------|
| Line voltage | 208-230 VAC | +/-10%                    |
| Low voltage  | 24 VAC      | +/-10%                    |
| DC bus       | 310 VDC     | Discharged before service |
| CN2 signal   | 24 VAC      | Call for fan              |
| Ambient      | see chart   | Record at test            |



If a code returns after clearing, inspect the harness before replacing the board.

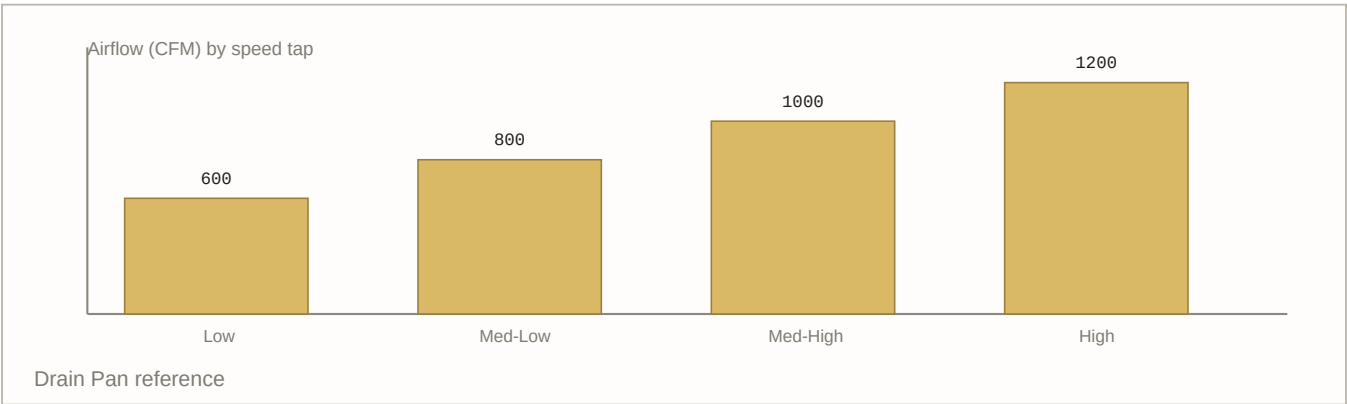
# 11. Drain Pan

The 38MURA uses a variable-speed inverter to match capacity to load. Because the inverter regulates motor speed, most faults present as a control-signal problem rather than a mechanical failure.

## Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

| Terminal | Wire | Function      |
|----------|------|---------------|
| CN2-1    | RED  | 24VAC supply  |
| CN2-2    | WHT  | Common return |
| CN2-3    | BLU  | Tach feedback |
| CN1-1    | BLK  | Earth ground  |
| CN1-2    | YEL  | Thermostat Y  |



Use the wiring diagram in Appendix D to confirm connector pinouts before probing.

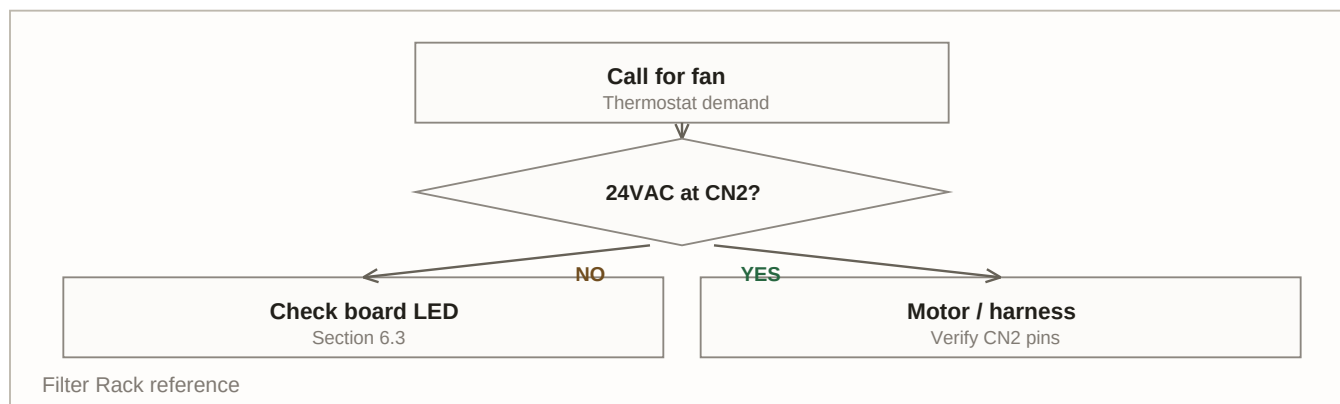
## 12. Filter Rack

Before opening any panel, disconnect power at the service switch and wait for the DC bus to discharge. Confirm 0VAC and 0VDC at the board before proceeding.

### Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

| Code | Meaning        | Action                   |
|------|----------------|--------------------------|
| E1   | High pressure  | Check charge and airflow |
| E2   | Low pressure   | Check charge and leak    |
| E4   | Locked rotor   | Check compressor         |
| E5   | AC overcurrent | See Section 5.2          |
| E6   | Communication  | Check harness            |



Always confirm 0VAC and 0VDC at the board before making resistance measurements.

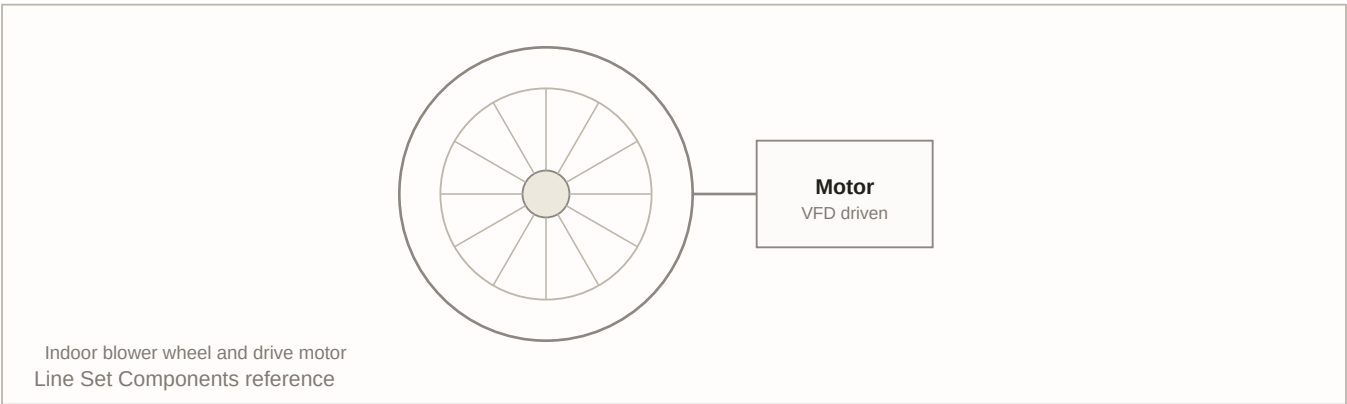
### 13. Line Set Components

Use a calibrated multimeter with a true-RMS function for all AC measurements. Resistance checks must be made with the unit de-energized and the harness disconnected.

**Procedure notes**

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
- Clear codes only after the repair is verified.

| Step            | Tool          | Spec      |
|-----------------|---------------|-----------|
| Mounting screws | 1/4 in driver | 2.5 N-m   |
| Terminals       | Torque driver | 0.8 N-m   |
| Line set        | Flare wrench  | per chart |
| Filter          | Hand          | snug      |
| Panel           | 1/4 in driver | 1.5 N-m   |



Record the outdoor ambient and the suction line temperature together for an accurate comparison.

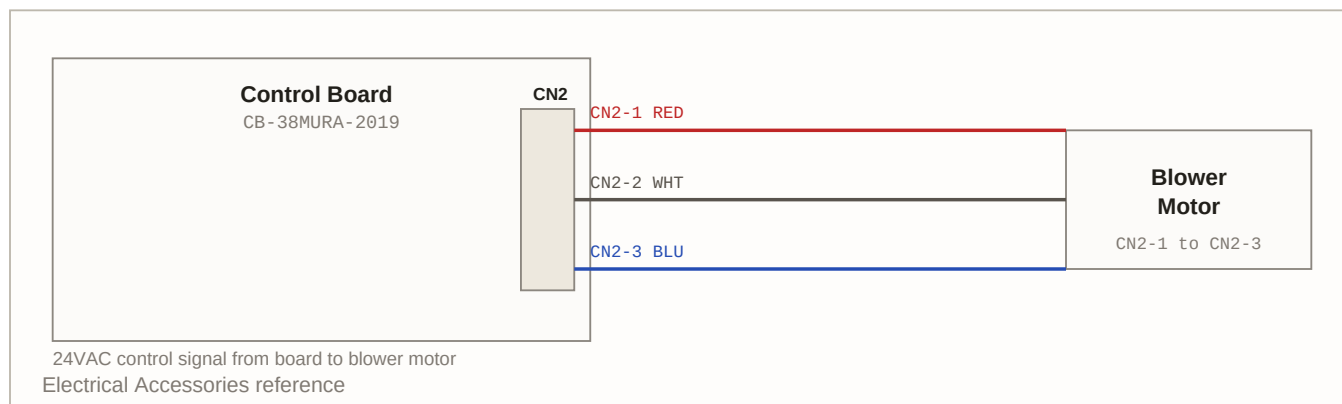
## 14. Electrical Accessories

Connector CN2 carries the low-voltage control signal between the board and the blower motor. A loose or corroded pin here is one of the most common causes of intermittent operation.

### Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

| Speed Tap | CFM  | Static     |
|-----------|------|------------|
| Low       | 600  | 0.20 in wc |
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If a code returns after clearing, inspect the harness before replacing the board.

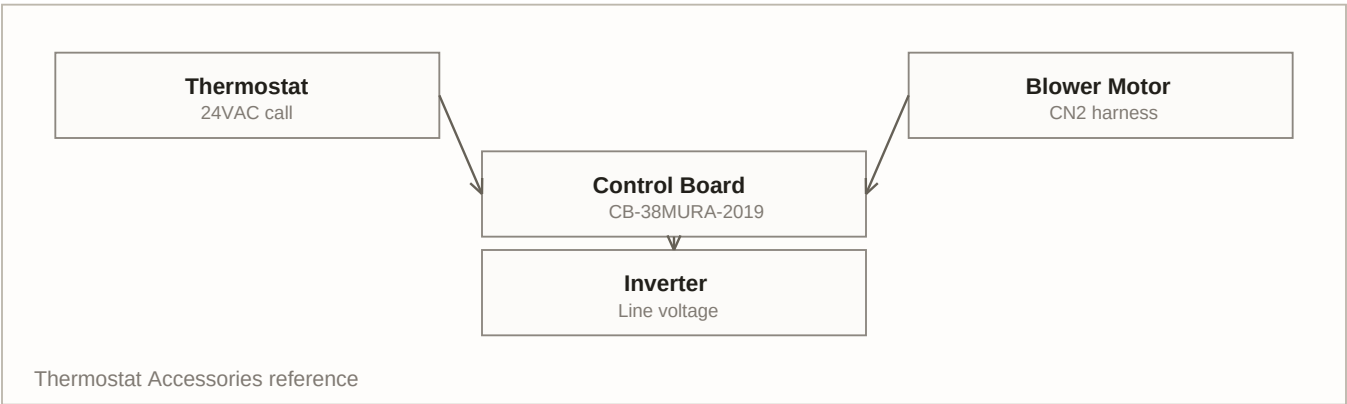
# 15. Thermostat Accessories

The board stores the last five fault codes in non-volatile memory. Review the stored codes before clearing them, because the history often points to the true root cause.

## Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize for 30 seconds.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not to an assumption.

| Item         | Expected    | Notes                     |
|--------------|-------------|---------------------------|
| Line voltage | 208-230 VAC | +/-10%                    |
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| DC bus       | 310 VDC     | Discharged before service |
| CN2 signal   | 24 VAC      | Call for fan              |
| Ambient      | see chart   | Record at test            |



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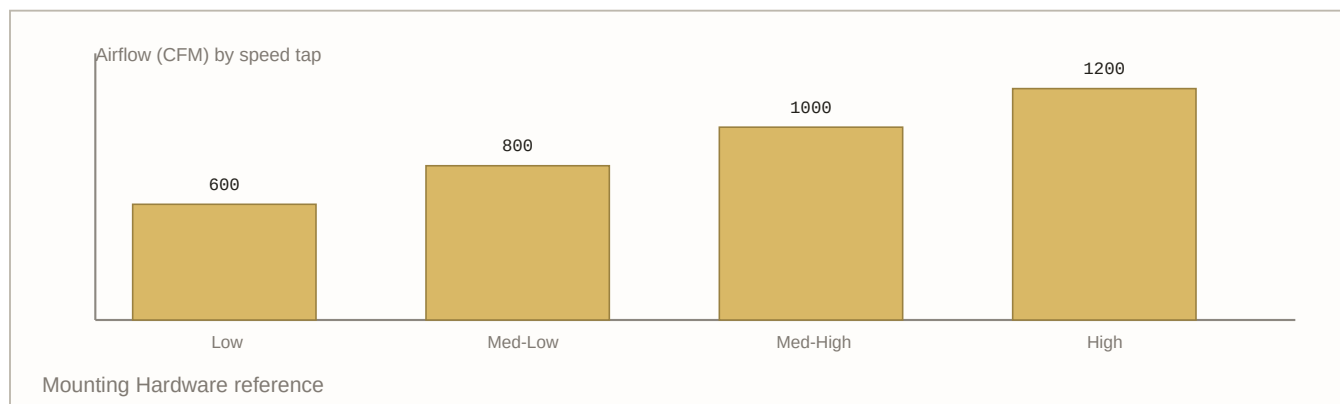
## 16. Mounting Hardware

Record ambient and supply-air temperatures at the same time so the temperature split can be compared against the charging chart for the current mode.

### Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
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| Terminal | Wire | Function      |
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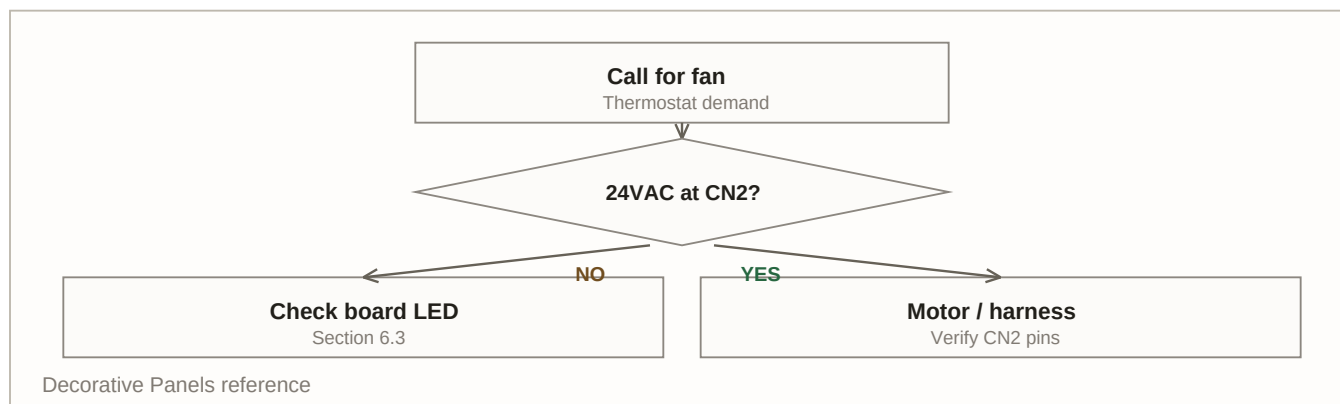
## 17. Decorative Panels

When a reading is out of range, do not replace the component until the associated wiring and connectors have been verified good. Most returned parts test good on the bench.

### Procedure notes

- Verify thermostat demand is present.
- Check the low-voltage transformer output.
- Measure the signal at CN2.
- Compare against the table in this section.
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| E6   | Communication  | Check harness            |



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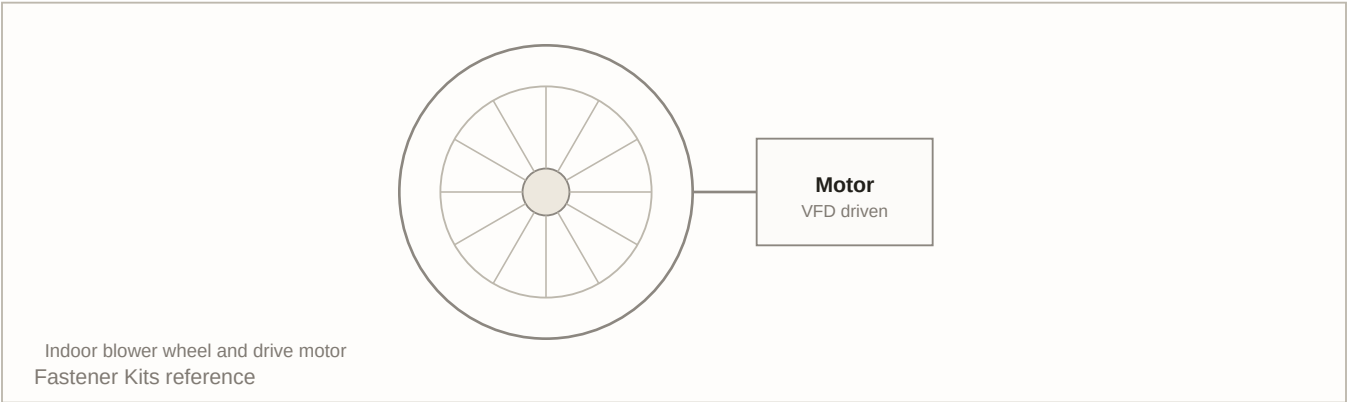
# 18. Fastener Kits

Torque all electrical terminations to the value shown in the specifications table. Loose terminations overheat and can damage the board terminals over time.

## Procedure notes

- Use only factory-approved replacement parts.
- Match the full part number, including revision.
- Transfer configuration jumpers where fitted.
- Confirm harness routing against the reference photos.
- Run a full startup sequence after assembly.

| Step            | Tool          | Spec      |
|-----------------|---------------|-----------|
| Mounting screws | 1/4 in driver | 2.5 N-m   |
| Terminals       | Torque driver | 0.8 N-m   |
| Line set        | Flare wrench  | per chart |
| Filter          | Hand          | snug      |
| Panel           | 1/4 in driver | 1.5 N-m   |



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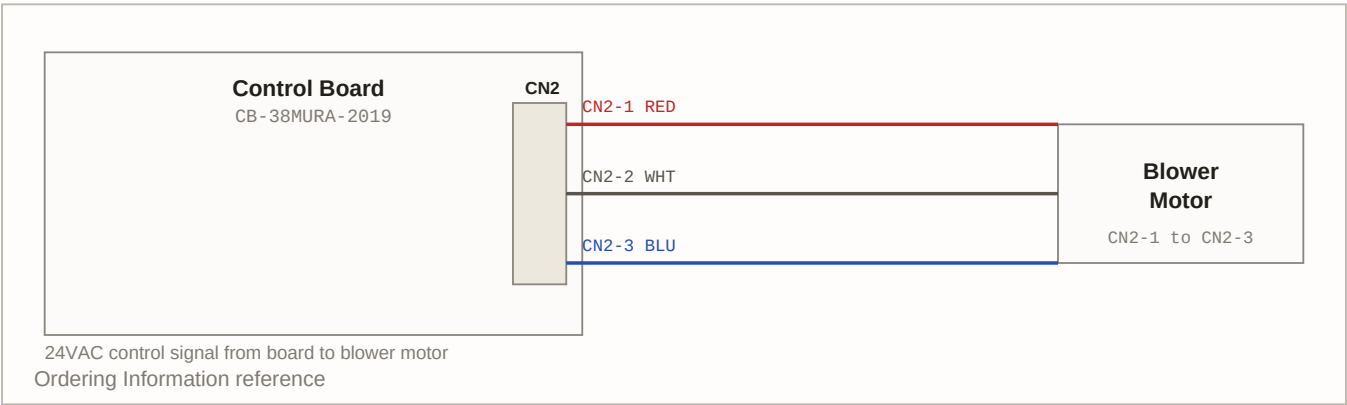
# 19. Ordering Information

Document the final readings and the parts used on the work order. Accurate records shorten the next visit and support the warranty claim.

## Procedure notes

- Measure with the unit in the same mode as the fault.
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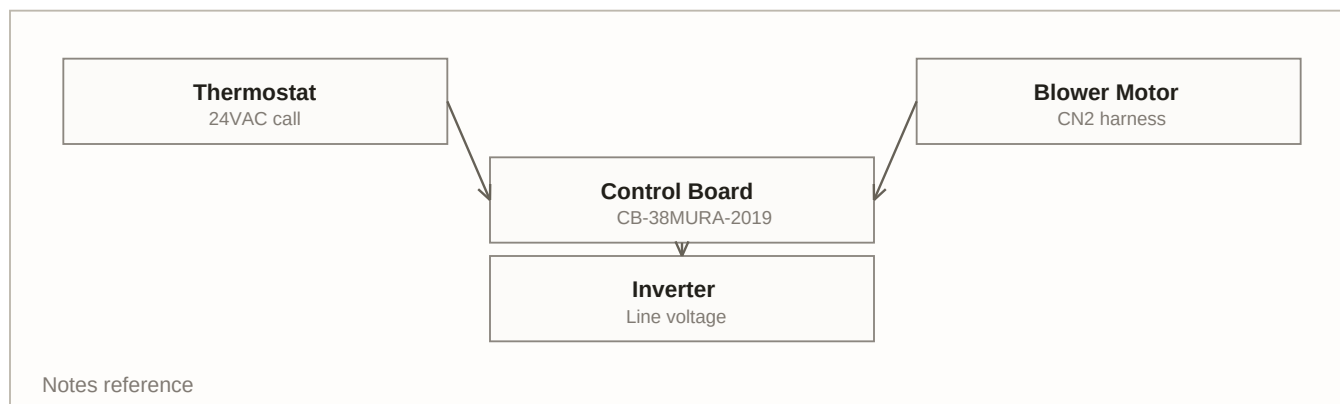
## 20. Notes

This section covers the recommended field procedure for the 2019 Carrier 38MURA. Follow the steps in order and verify each measurement before moving to the next step. Record every reading on the service worksheet.

### Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC at the board.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

| Item         | Expected    | Notes                     |
|--------------|-------------|---------------------------|
| Line voltage | 208-230 VAC | +/-10%                    |
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| DC bus       | 310 VDC     | Discharged before service |
| CN2 signal   | 24 VAC      | Call for fan              |
| Ambient      | see chart   | Record at test            |



Always confirm 0VAC and 0VDC at the board before making resistance measurements.